Southern New Hampshire University

CS 499 Module One Assignment Template

Complete this template by replacing the bracketed text with the relevant information.

I. **Self-Introduction:** Address all of the following questions to introduce yourself.

A. How long have you been in the Computer Science program?
I started college in 2019 at Kellogg Community College doing my general education credits. I was using something called the 'Legacy Promise' scholarship, which was 2 free years of college. After I used that I took a break for 2 years to get money for funding. In July of 2024 I started the computer science program at Southern New Hampshire University. So a year and a half to 2 years.

B. What have you learned while in the program? List three of the most important concepts or skills you have learned.
I've learned a lot of new skills I didn't think I would. I knew I would learn various languages like Java, python, and C++. I didn't know I would be learning anything about SQL or Mongodb. It was interesting to learn and showed me how important it is to the computer science field. Learning to have versatility and diversity in your code is important. Another skill I've learned is organized writing and how it influences technical writing. It is important to be a strong writer in computer science. The ability to communicate clearly and in a manner that the audience would understand is truly important. The last key skill I've learned is problem solving and collaboration. It is important to think outside the box or ask for help when dealing with a new challenge that is not easily understood. Being able to collaborate with others in various methods is also extremely important, whether this be waterfall, agile, or something different entirely. The ability to be flexible is extremely important in this field.

C. Discuss the specific skills you aim to demonstrate through your enhancements to reach each of the course outcomes.
I wish to demonstrate my creativity. I've felt one of my best skills is the ability to think outside the box, or look at the opposite perspective from someone else. Along with this I want to demonstrate my ability to solve problems. Most organizations depend on computer scientists to solve problems or make a certain task easier. I also wish to demonstrate my adaptability. It is important to be adaptable in this industry as it is never certain what problem may arise, or what different technology or software may be needed.

D. How do the specific skills you will demonstrate align with your career plans related to your degree?
These skills align with my career plans as I want to become a penetration tester and a security posture consultant. This role requires someone to be creative on how to penetrate a system or structure. You need to be a great problem solver in order to think of ways to get in or to think of ways to defend attack vectors. Adaptability is the most important as the cyber landscape is always changing. New attack vectors and new exploits arise every day.

E. How does this contribute to the specialization you are targeting for your career?

These enhancements contribute directly to my chosen specialization by helping me refine my technical and creative skills. I'm pursing a path that combined cyber security and software design, so improving my ability to think critically, solve problems, develop secure and efficient systems aligns closely with my professional goals. By revisiting past artifacts I can apply new techniques, strengthen weak areas, and demonstrate a deeper understanding of ethical and functional design.

II. ePortfolio Set Up:

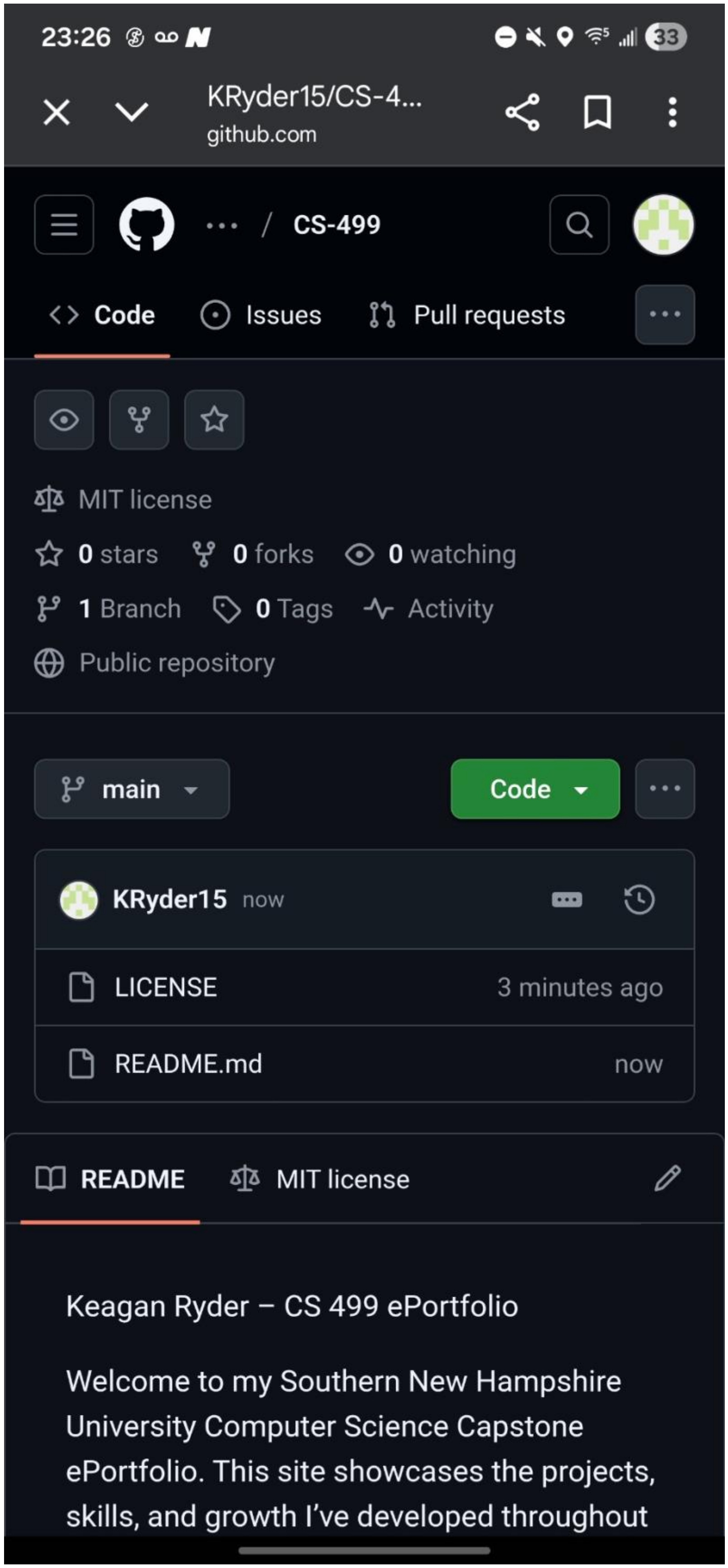
A. Submit a **screen capture** of your ePortfolio GitHub Pages home page that clearly shows your URL.
   i. You already have a repository in GitHub where you uploaded projects in previous courses. Your ePortfolio will reside in GitHub but can link to work at other sites, such as Bitbucket.

B. Use the GitHub Pages link in the Resource section for directions on:
   i. How to create your GitHub website and publish code to GitHub Pages
   Issues, such as adding links to other sites

C. Paste a screenshot of your GitHub Pages home page with your URL clearly showing in the space below.

URL: https://github.com/KRyder15/CS-499

Southern New Hampshire University



III. Enhancement Plan:

A. **Category One:** Software Engineering and Design

i. **Select an artifact** that is **aligned with the** software engineering and design **category** and explain its origin. Submit a file containing the code for the artifact you choose with your enhancement plan.

I will use my CS 360 Android Inventory Management App as my artifact for this category. This project was originally developed in Java using Android Studio to demonstrate software engineering and design principles in a mobile environment. The app allows users to securely log in, add, edit, delete, and view inventory items, and send SMS notifications when stock levels are low. It utilizes SQLite for local data storage, Activities and Fragments for modular UI design, and follows an organized Model-View-Controller architecture to maintain clean separation between data handling, logic, and user interface components.

The project demonstrates key software engineering skills such as code modularity, version control, exception handling, and UI responsiveness. It also reflects thoughtful design decisions, including usability, data validation, and maintainability. This artifact aligns with the Software Engineering and Design category because it showcases how structured design principles and software architecture can be applied to develop an interactive, reliable, and user-friendly application.

Note: Your artifact may be work from the following courses:

- IT 145: Foundation in Application Development
- CS 250: Software Development Lifecycle
- CS 260: Data Structures and Algorithms
- IT 315: Object Oriented Analysis and Design
- CS 320: Software Testing, Automation, and Quality Assurance
- CS 330: Computational Graphics and Visualization
- CS 340: Advanced Programming Concepts
- CS 350: Emerging Systems Architectures and Technologies
- CS 360: Mobile Architecture and Programming
- IT 365: Operating Environments
- IT 380: Cybersecurity and Information Assurance
- CS 405: Secure Coding
- CS 410: Reverse Software engineering
- IT 340: Network and Telecommunication Management
- IT 380: Cybersecurity and Information Assurance

ii. **Describe** a practical, well-illustrated **plan** for enhancement in alignment with the category, including a pseudocode or flowchart that illustrates the planned enhancement.

Enhancement plan

Goal

Refactor the Java Android Inventory Management App to improve architecture, reliability, and usability by adding an offline-first sync capability with a cloud API, migrating SQLite access to Room with a repository layer, introducing WorkManager based background sync, improving accessibility, and adding automated tests.

Scope

1. Migrate direct SQLite operations to Room with entities, DAO interfaces, and migrations
2. Introduce a Repository that mediates between Room and a Retrofit based REST API
3. Add WorkManager periodic and on-demand sync with conflict resolution using updatedAt timestamps
4. Improve UI responsiveness and accessibility through content descriptions, touch target sizing, and basic theming for contrast
5. Add unit tests for view models and repository, plus instrumented UI tests for critical flows
6. Secure local secrets using EncryptedSharedPreferences and validate inputs to reduce crashes

Deliverables

Refactored codebase with Room, Repository, Retrofit, and WorkManager
Updated layouts and accessibility attributes
Test suite with coverage reports and a short architecture readme

Pseudocode

```
OnAppStart():
  initDependencies()
  if userLoggedIn():
    WorkManager.schedulePeriodicSync()
  navigateToMain()

AddItemFlow(input):
  if validate(input) == false:
    ui.showError("Invalid fields")
    return
  item = Item.from(input).withUpdatedAt(now())
  repository.saveLocal(item)
  ui.showToast("Saved locally")
  WorkManager.enqueueSyncNow()

Repository.saveLocal(item):
  roomDao.upsert(item)
  markPending(item.id)
```

```
Repository.sync():
  localChanges = roomDao.getPendingChanges()
  for each change in localChanges:
    serverVersion = api.getItem(change.id)
    if serverVersion == null:
      api.create(change)
      roomDao.markSynced(change.id)
    else if serverVersion.updatedAt > change.updatedAt:
      // server wins
      roomDao.replace(serverVersion)
      roomDao.markSynced(serverVersion.id)
    else:
      api.update(change.id, change)
      roomDao.markSynced(change.id)

  deltas = api.fetchDelta(since = lastSyncTime())
  roomDao.merge(deltas)
  setLastSyncTime(now())

EditItemFlow(id, editedInput):
  if validate(editedInput) == false:
    ui.showError("Invalid fields")
    return
  item = repository.findLocal(id)
  item.apply(editedInput).withUpdatedAt(now())
  repository.saveLocal(item)
  WorkManager.enqueueSyncNow()

DeleteItemFlow(id):
  repository.softDeleteLocal(id)      // mark deleted, keep tombstone
  WorkManager.enqueueSyncNow()

Repository.softDeleteLocal(id):
  item = roomDao.get(id)
  item.deleted = true
  item.updatedAt = now()
  roomDao.upsert(item)
  markPending(id)

PeriodicSyncWorker.doWork():
  Repository.sync()
  return Success

validate(input):
  return input.name not empty
    and input.quantity >= 0
    and input.sku matches "[A-Z0-9-]{4,}"
```

Flowchart description

App start
arrow to dependency initialization
diamond user logged in
yes path schedules periodic sync and opens main screen
no path opens login then returns to main on success

User creates or edits an item
process validate input
diamond valid
no path shows error and returns to form
yes path writes to Room through repository and marks pending
process UI shows local commit feedback
process WorkManager enqueues immediate sync
process Sync worker pushes local pending changes, fetches server deltas, merges into Room, updates last sync time
UI observes database changes and updates list detail screens

Testing plan

Unit tests
view model validation and state transitions
repository conflict resolution with mocked API and in-memory Room

Instrumented tests
add edit delete item flows with Espresso
verify accessibility labels exist and importantForAccessibility is set for key controls

Security and reliability checks

Store tokens with EncryptedSharedPreferences
sanitize user input and guard database calls with try catch and meaningful error messages
use Retrofit timeouts and exponential backoff in the sync worker

Success criteria

All CRUD flows work offline and reconcile within one minute of connectivity
No accessibility violation reported by Accessibility Scanner on primary screens
At least eighty percent line coverage on view models and repository tests
Sync conflicts resolve deterministically using updatedAt semantics

For this category of enhancement, consider improving a piece of software, transferring a project into a different language, reverse engineering a piece of software for a different operating system, or expanding a project's complexity. These are just recommendations. Consider being creative and proposing an alternative enhancement to your instructor.

Think about what additions to include to complete the enhancement criteria in this category. Since one example option is to port to a new language, that is the kind of scale that is expected. This does not mean you need to port to a new language but instead have an equivalent scale of enhancement. Underlying expectations of any enhancement include fixing errors, debugging, and cleaning up comments, but these are not enhancements themselves.

iii. Explain how the planned enhancement will **demonstrate** specific **skills** and align with course outcomes.

a. Identify and describe the specific skills you will demonstrate that align with the course outcome.

The planned enhancement will demonstrate my ability to apply software engineering principles such as modular design, scalability, and maintainability in a real-world application. By refactoring the app to use Room, a Repository layer, and WorkManager for background synchronization, I will show skills in software architecture design, API integration, and asynchronous processing. Implementing accessibility improvements will demonstrate my understanding of inclusive design and user-centered development, while writing unit and UI tests will highlight my skills in automated testing and quality assurance. Together, these enhancements show that I can design, build, and refine professional-quality software that is reliable, maintainable, and user friendly.

b. Select one or more of the course outcomes below that your enhancement will align with.

This enhancement aligns with the following course outcomes:

Outcome 3: Design and evaluate computing solutions that solve a given problem using algorithmic principles and computer science practices and standards appropriate to its solution while managing the trade-offs involved in design choices.

Outcome 4: Demonstrate an ability to use well-founded and innovative techniques, skills, and tools in computing practices for the purpose of implementing computer solutions that deliver value and accomplish industry-specific goals.

Outcome 5: Develop a security mindset that anticipates adversarial exploits in software architecture and designs to expose potential vulnerabilities, mitigate design flaws, and ensure privacy and enhanced security of data and resources.

Course Outcomes:

1. Employ strategies for building collaborative environments that enable diverse audiences to support organizational decision-making in the field of computer science.
2. Design, develop, and deliver professional-quality oral, written, and visual communications that are coherent, technically sound, and appropriately adapted to specific audiences and contexts.
3. Design and evaluate computing solutions that solve a given problem using algorithmic principles and computer science practices and standards appropriate to its solution while managing the trade-offs involved in design choices.
4. Demonstrate an ability to use well-founded and innovative techniques, skills, and tools in computing practices for the purpose of implementing computer solutions that deliver value and accomplish industry-specific goals.
5. Develop a security mindset that anticipates adversarial exploits in software architecture and designs to expose potential vulnerabilities, mitigate design flaws, and ensure privacy and enhanced security of data and resources.

B. **Category Two:** Algorithms and Data Structures

i. **Select an artifact** that is **aligned with the** algorithms and data structures **category** and explain its origin. Submit a file containing the code for the artifact you choose with your enhancement plan. You may choose work from the courses listed under Category One.

I will use my CS 300 auction bids project as the artifact for this category. The original assignment implemented a singly linked list in C++ to store and manage bid records loaded from a CSV file, with operations to insert, remove, search by id, and traverse the list to print results. The program included a simple menu driven interface, a Bid struct, and ListNode classes that maintained head and tail pointers. It emphasized pointer manipulation, dynamic memory management, and linear time operations characteristic of a linked list. This artifact fits the algorithms and data structures category because it demonstrates concrete use of an abstract data type, algorithmic operations over that structure, and analysis of time and space tradeoffs in insertion, search, and deletion. I will submit the source files including the Bid structure definition, the linked list implementation with insert remove search and print methods, the CSV loader utility, and the main program used to run tests.

ii. **Describe a practical, well-illustrated plan for enhancement in alignment with the category, including a pseudocode or flowchart that illustrates the planned enhancement.**

Enhancement plan

Goal
Upgrade the auction bids program by replacing the singly linked list with two data structures: an AVL tree keyed by bidId for fast search, insert, and delete, and a min heap keyed by amount for O(1) access to the lowest bid. Add a benchmarking harness to compare performance with the original list.

Scope

1. Implement AVL tree (insert, search, delete, in-order traversal) keyed by bidId
2. Implement min heap keyed by (amount, bidId) with lazy delete via tombstones
3. Maintain cross references so deletes update both structures
4. Add CSV loader that populates both structures
5. Add benchmark mode to measure operations across list vs AVL+heap
6. Output summary tables (n, ops/sec, avg ns/op)

Pseudocode

Data types Bid { id, title, fund, amount } AvlNode { key=id, bid, height, left, right } HeapNode { key=(amount, id), bidId, tombstone }

InsertBid(bid) avl.insert(bid.id, bid) heap.push(HeapNode(key=(bid.amount, bid.id), bidId=bid.id, tombstone=false))

FindBid(id) return avl.search(id) // O(log n)

RemoveBid(id) bid = avl.search(id) if bid == null: return false avl.delete(id) // O(log n) heap.markTombstone(id) // O(1) in hash map of id->heap index or O(log n) without map return true

GetLowestBid() while heap.notEmpty(): top = heap.top() if top.tombstone == true: heap.pop() continue return getBidById(top.bidId) // avl.search(top.bidId) return null

LoadCSV(path) for row in csv(path): bid = parseBid(row) InsertBid(bid)

Benchmark() sizes = [1e3, 5e3, 1e4, 5e4] for n in sizes: bids = generateNBids(n) timeList = timeOpsUsingLinkedList(bids) timeAvlHeap = timeOpsUsingAvlAndHeap(bids) printSummary(n, timeList, timeAvlHeap)

timeOpsUsingAvlAndHeap(bids) start = now() for b in bids: InsertBid(b) tInsert = now() - start start = now() for b in sampleIds(bids, k=1000): FindBid(b.id) tSearch = now() - start start = now() for b in sampleIds(bids, k=1000): RemoveBid(b.id) tDelete = now() - start return {insert:tInsert, search:tSearch, delete:tDelete}

AVL insert (sketch) insert(node, key, bid): if node == null: return new AvlNode(key, bid) if key < node.key: node.left = insert(node.left, key, bid) else if key > node.key: node.right = insert(node.right, key, bid) else: node.bid = bid; return node updateHeight(node) balance = height(node.left) - height(node.right) if balance > 1 and key < node.left.key: return rotateRight(node) if balance < -1 and key > node.right.key: return rotateLeft(node) if balance > 1 and key > node.left.key: node.left = rotateLeft(node.left); return rotateRight(node) if balance < -1 and key < node.right.key: node.right = rotateRight(node.right); return rotateLeft(node) return node

Heap operations (sketch) push(node): heap.append(node) bubbleUp(lastIndex) top(): return heap[0] pop(): swap(0, last) remove last bubbleDown(0) markTombstone(id): idx = idToHeapIndex.get(id) // optional optimization if idx exists: heap[idx].tombstone = true else: tombstoneMap.add(id) // fall back; check in GetLowestBid

Flowchart description

Start
choose mode from menu: Load CSV, Insert, Find, Remove, Lowest, Benchmark, Quit
Load CSV
parse each row
InsertBid into AVL and Heap
Insert
validate input
InsertBid
Find
FindBid via AVL
Remove
RemoveBid, mark heap tombstone
Lowest
GetLowestBid using heap and confirm with AVL
Benchmark
run timing suite and print table
Quit

Success criteria

Search, insert, delete average O(log n) with AVL
GetLowestBid amortized O(1) using heap
Benchmark shows improved times versus the singly linked list for n ≥ 5,000
Functional parity preserved with original features (load, add, find, remove, print)

For this category of enhancement, consider improving the efficiency of a project or expanding the complexity of the use of data structures and algorithms for your artifact. These are just recommendations. Consider being creative and proposing an alternative enhancement to your instructor. Note: You only need to choose one type of enhancement per category.

Think about what additions to include to complete the enhancement criteria in this category. Since one example option is to port to a new language, that is the kind of scale that is expected. Perhaps you might increase the efficiency and time complexity of an algorithm in an application and detail the logic of the increased time complexity. Remember, you do not need to port to a new language but instead have an equivalent scale of enhancement. Underlying expectations of any enhancement include fixing errors, debugging, and cleaning up comments, but these are not enhancements themselves.

iii. Explain how the planned enhancement will **demonstrate** specific **skills** and align with course outcomes.

a. Identify and describe the specific skills you will demonstrate to align with the course outcome.

I will demonstrate skills in algorithm design, data structure implementation, and performance optimization by enhancing the program to use an AVL tree and a min heap instead of a singly linked list. This will show my ability to apply efficient algorithms, analyze time complexity, and use Big O notation to justify design choices. I will also demonstrate debugging and testing skills by verifying that the new structures maintain correct functionality while improving search, insert, and delete performance.

b. Select one or more of the course outcomes listed under Category One that your enhancement will align with.

This enhancement aligns with the following course outcomes:

Outcome 3: Design and evaluate computing solutions that solve a given problem using algorithmic principles and computer science practices and standards appropriate to its solution while managing the trade-offs involved in design choices.

Outcome 4: Demonstrate an ability to use well-founded and innovative techniques, skills, and tools in computing practices for the purpose of implementing computer solutions that deliver value and accomplish industry-specific goals.

C. **Category Three: Databases**

i. **Select an artifact** that is **aligned with the** databases **category** and explain its origin. Submit a file containing the code for the artifact you choose with your enhancement plan. You may choose work from the courses listed under Category One.

I will use my CS 340 Client/Server Development CRUD project as the artifact for this category. It originated as a Python application that connects to MongoDB using PyMongo and provides create, read, update, and delete operations through a reusable CRUD module. The project also includes a Dash dashboard for visualizing and filtering records, with pages that list items, show details, and render basic charts from aggregation queries. The codebase demonstrates data modeling choices for MongoDB collections, input validation, and use of indexes to support common queries. I will submit the source files including the CRUD module, the Dash app, configuration file, and example notebooks or scripts used to populate and query the database.

ii. **Describe** a practical, well-illustrated **plan** for enhancement in alignment with the category, including a pseudocode or flowchart that illustrates the planned enhancement.

Enhancement plan

Goal
Strengthen the CS 340 CRUD project by adding schema validation, indexed queries, secure REST APIs, caching, and richer analytics while keeping the Dash dashboard in sync.

Scope

1. Add MongoDB JSON schema validation and Pydantic request models

2. Create compound indexes guided by explain plans

3. Expose a FastAPI REST layer with JWT authentication and role based access

4. Add Redis caching for frequent read queries with cache invalidation on writes

5. Expand analytics using aggregation pipelines and integrate results into the dashboard

6. Write unit tests for CRUD and aggregation logic and smoke tests for the API

Pseudocode

Startup load_config() mongo = connect() ensure_indexes(mongo) ensure_schema_validators(mongo) cache = connect_redis()

Model class ItemModel: id name state createdAt updatedAt deleted

POST /api/items require_role(editor) payload = validate_with_pydantic(request.body) payload.updatedAt = now() mongo.items.insert_one(payload) cache.invalidate(prefix=”items:”) return 201

GET /api/items?state=MI&page=1 key = “items:list:state=MI:page=1” if cache.has(key): return cache.get(key) cursor = mongo.items.find({state:”MI”, deleted:false}) .sort({createdAt:-1}) .skip(50*(page-1)).limit(50) results = list(cursor) cache.set(key, results, ttl=300) return results

PATCH /api/items/{id} require_role(editor) update = validate_with_pydantic(request.body) update.updatedAt = now() mongo.items.update_one({_id:id}, {“$set”: update}) cache.invalidate(prefix=f”items:{id}”) cache.invalidate(prefix=”items:list”) return 200

DELETE /api/items/{id} require_role(editor) mongo.items.update_one({_id:id}, {“$set”: {deleted:true, updatedAt:now()}}) cache.invalidate(prefix=f”items:{id}”) cache.invalidate(prefix=”items:list”) return 204

Aggregation example counts by state pipeline = [{“$match”: {“country”:”United States”, “deleted”:false}}, {“$group”: {“_id”:”$state”, “count”:{“$sum”:1}}}, {“$sort”: {“count”:-1}}] results = mongo.items.aggregate(pipeline)

Indexes mongo.items.create_index([(“state”,1), (“createdAt”,-1)]) mongo.items.create_index([(“name”,1)]) mongo.items.create_index([(“deleted”,1)])

Dash integration dashboard calls GET endpoints dashboard charts call analytics endpoint that runs aggregation cache warms popular queries on startup

Flowchart description

Client request
Arrow to FastAPI router
Diamond auth and role check
No path returns 401 or 403
Yes path goes to handler
Handler validates input
Diamond cache hit for reads
Yes path returns cached data
No path queries MongoDB with indexes
Optional aggregation runs
Handler stores result in cache for reads
Writes invalidate related cache keys
Response returned to client and dashboard updates

Success criteria

Schema validation blocks invalid writes
Indexed queries reduce read latency on filtered lists
JWT roles restrict create update delete to authorized users
Caching lowers response time on repeated reads
Analytics endpoints deliver state level and time series insights
Automated tests pass for CRUD and aggregation paths

For this category of enhancement, consider adding more advanced concepts of MySQL, incorporating data mining, creating a MongoDB interface with HTML/JavaScript, or building a full stack with a different programming language for your artifact. These are just recommendations; consider being creative and proposing an alternative enhancement to your instructor. Note: You only need to choose one type of enhancement per category.

Think about what additions to include to complete the enhancement criteria in this category. Since one example option is to port to a new language, that is the kind of scale that is expected. Perhaps you might increase the efficiency and time complexity of an algorithm in an application and detail the logic of the increased time complexity. Remember, you do not need to port to a new language but instead have an equivalent scale of enhancement. Underlying expectations of any enhancement include fixing errors, debugging, and cleaning up comments, but these are not enhancements themselves.

iii. Explain how the planned enhancement will **demonstrate** specific **skills** and align with course outcomes.

a. Identify and describe the specific skills you will demonstrate that align with the course outcome.

I will demonstrate the following skills that align with the databases category:

Data modeling and validation
Design MongoDB collection schemas and enforce correctness using JSON Schema and Pydantic models to prevent invalid writes and maintain data integrity.

Query optimization
Create and tune single-field and compound indexes based on explain plans, measure query latency, and justify index choices with before and after metrics.

API design and integration
Build a secure REST layer with FastAPI that cleanly exposes CRUD and analytics endpoints, handle pagination and filtering, and integrate the API with the dashboard client.

Authentication and authorization
Implement JWT based auth and role based access control so only authorized roles can create, update, or delete records, demonstrating least privilege and secure session handling.

Caching strategy
Use Redis to cache frequent read queries and implement precise cache invalidation on writes to balance freshness with performance.

Analytics with aggregation pipelines
Author MongoDB aggregation pipelines for counts, trends, and group-by summaries and expose them to the dashboard for visual insights.

Testing and reliability
Write unit tests for CRUD and aggregation logic, and smoke tests for API endpoints to ensure correctness and prevent regressions.

Operational readiness
Add environment based configuration, connection pooling, and basic monitoring logs to support deployment and troubleshooting.

b. Select one or more of the course outcomes listed under Category One that your enhancement will align with.

This enhancement aligns with the following course outcomes:

Outcome 3: Design and evaluate computing solutions that solve a given problem using algorithmic principles and computer science practices and standards appropriate to its solution while managing the trade-offs involved in design choices.

Outcome 4: Demonstrate an ability to use well-founded and innovative techniques, skills, and tools in computing practices for the purpose of implementing computer solutions that deliver value and accomplish industry-specific goals.

Outcome 5: Develop a security mindset that anticipates adversarial exploits in software architecture and designs to expose potential vulnerabilities, mitigate design flaws, and ensure privacy and enhanced security of data and resources.

IV. ePortfolio Overall Skill Set

A. Accurately describe the **skill set** to be illustrated by the **ePortfolio overall**.

i. Skills and outcomes planned to be illustrated in the code review

The ePortfolio will illustrate my overall skill set as a computer science professional by showcasing my ability to design, develop, and refine software solutions across multiple domains. In the code review, I plan to demonstrate skills in software engineering and architecture, algorithm design and analysis, and database integration. This includes showing how I apply structured design principles, write efficient and maintainable code, and use data-driven approaches to improve performance and reliability. The review will highlight my understanding of modular programming, problem solving, and security-minded development while aligning with key outcomes such as designing computing solutions, applying innovative tools and techniques, and ensuring system integrity and user accessibility.

ii. Skills and outcomes planned to be illustrated in the narratives

In the narratives, I plan to illustrate my ability to reflect on my development process, explain design decisions, and connect technical work to real-world applications. I will demonstrate strong communication skills by clearly describing the purpose, improvements, and impact of each enhancement. The narratives will show my growth in areas such as problem solving, critical thinking, and ethical computing practices. They will also highlight how I've applied course outcomes—designing effective computing solutions, employing innovative techniques, and maintaining a security-focused mindset—to produce professional, well-documented, and user-centered software projects.

iii. Skills and outcomes planned to be illustrated in the professional self-assessment

In the professional self-assessment, I plan to illustrate my ability to evaluate my strengths, growth, and readiness to apply computer science principles in a professional environment. I will demonstrate self-awareness by reflecting on how my coursework and projects have developed my technical and interpersonal skills, including software design, problem solving, collaboration, and communication. This section will connect my academic experiences to industry expectations, showing how I can adapt to new technologies, think critically about design choices, and approach challenges with a security-

focused and ethical mindset. It will also highlight how I plan to continue learning and improving as a computing professional.